

will get a bell curve. Figure 6 shows a bell curve (courtesy Wikipedia). The exact shape of the bell curve depends on the mean and standard deviation. But what does this curve signify? Let us try to understand this by analysing a natural phenomenon. A cursory search on the internet told me that the average height of an Indian man now is 1.73 metres (5 feet and 8 inches). Now, if you look around, you will see a lot of men with height very close to this figure. The chances of you observing a man who has a height below 1.42 metres (4 feet and 8 inches) or above 2 metres (6 feet and 8 inches) are very slim. Now, you go out and record the heights of say 1,000,000 men and plot a graph. Let the x-axis (horizontal axis) mark the height of a person and the y-axis (vertical axis) mark the number of samples analysed (in this case heights of 1,000,000 people). When you plot this graph, you will see a bell curve, with a few minor skews and bends. So, the mathematical equation that represents a normal distribution can easily capture natural phenomena like the heights of human beings.

Now, let us see an example for using normal distribution. Consider the code shown in Figure 7. We use Matplotlib for plotting and NumPy for the function `normal()` which implements normal distribution. From Line 3, we can see that the sample size is 1000. Line 4 tells us that we are plotting a histogram with 1000 bins. However, if you observe the bell curve in Figure 7, this is not what we expected. This bell curve doesn't have much similarity with the one we saw in Figure 6. What is the reason for this? Remember, our sample size is just 1000. This is like an election exit-poll where you ask the opinion of just 100 people and then make a prediction. The sample size should be sufficiently large to get a clearer picture. Figure 8 shows a bell curve when line 3 in the code in Figure 7 is replaced with the line '`sample = normal(size=100000000)`' and the program is executed again. This

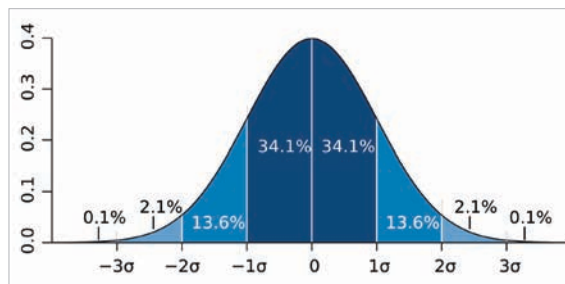


Figure 6: A bell curve (Courtesy: Wikipedia)

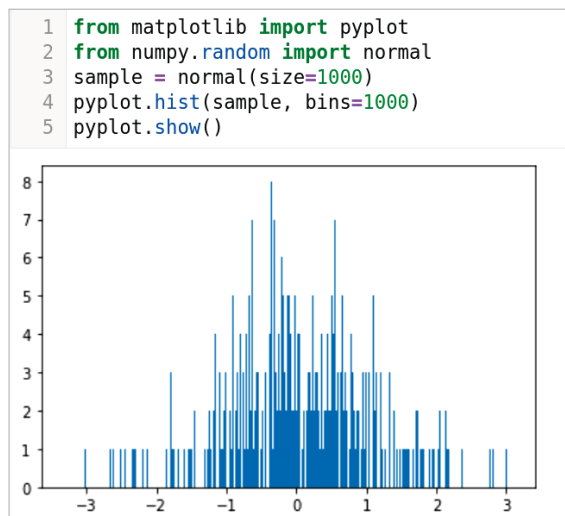


Figure 7: Plotting a bell curve

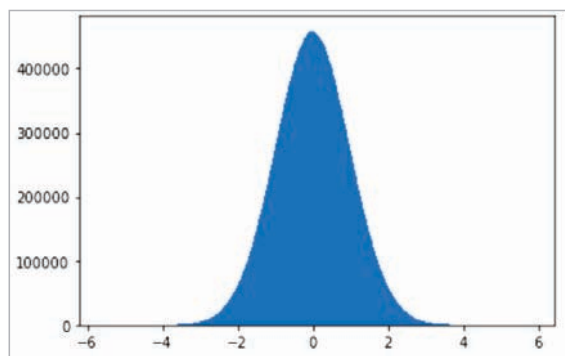


Figure 8: A better bell curve

time our sample size is 100,000,000 and the bell curve looks almost like the one shown in Figure 6. Normal distributions and bell curves are just the beginning.

a healthy and tasty soup (AI based applications). From the next article onwards, we will begin cooking that soup. **END** 🐧

In the next article in this series we will discuss other probability distribution functions that can represent and generalise many other events and natural phenomena. Next time, we will also address the topic a bit more formally.

Now it is time for us to wind up our discussion for the time being. In the next article in this series, we will continue exploring concepts in probability and statistics. We will also install and use Anaconda, a distribution of Python for scientific computing, which is especially useful for developing AI, machine learning and data science based applications. As mentioned earlier, we will also get familiar with yet another Python library called Theano, which is used a lot in the fields of AI and machine learning. But it is also high time to do one more thing in this series. So far, we were collecting vegetables (Python libraries) and spices (mathematical concepts) to make

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